FDN benchmark v1: verified MAE comparison

PEMS04

	Seed 42	Seed 1	Seed 2	Mean ± SD
GWN	19.0589	19.0386	19.1735	19.0903 ± 0.0727
DualGWN	18.7365	**18.6852**	18.9050	18.7755 ± 0.1150
FDN fixed-K3	19.0789	18.9649	18.8235	18.9558 ± 0.1279
FDN learnable-K3	**18.6643**	18.7734	**18.8095**	**18.7491 ± 0.0756**

Bold outline marks the lowest MAE in each column.

PEMS08

	Seed 42	Seed 1	Seed 2	Mean ± SD
GWN	14.7284	**14.7476**	14.9529	**14.8096 ± 0.1244**
DualGWN	**14.6566**	15.1148	15.0230	14.9315 ± 0.2424
FDN fixed-K3	14.6709	15.0596	**14.7286**	14.8197 ± 0.2098
FDN learnable-K3	14.9993	15.1722	14.8936	15.0217 ± 0.1406

Bold outline marks the lowest MAE in each column.

Component effects: ΔMAE

	PEMS04	PEMS08
DualGWN - GWN	**-0.3148** **3/3 seeds better** **better on mean**	**+0.1219** **1/3 seeds better** **worse on mean**
fixed-K3 - DualGWN	**+0.1802** **1/3 seeds better** **worse on mean**	**-0.1117** **2/3 seeds better** **better on mean**
learnable-K3 - fixed-K3	**-0.2067** **3/3 seeds better** **better on mean**	**+0.2020** **0/3 seeds better** **worse on mean**
learnable-K3 - DualGWN	**-0.0265** **2/3 seeds better** **better on mean**	**+0.0902** **1/3 seeds better** **worse on mean**
learnable-K3 - GWN	**-0.3412** **3/3 seeds better** **better on mean**	**+0.2121** **1/3 seeds better** **worse on mean**

Δ = left model − right model. Negative values are better.

Pre-registered Go / No-Go

Dataset means < 0	**1 / 2**	required 2 / 2
Negative seed pairs	**3 / 6**	required ≥ 5 / 6
CI upper bound < 0	**0 / 2**	required ≥ 1 / 2

STOP independent method paper

The decision follows the frozen threshold; no criterion passed.

24 runs · 3 training seeds · 100 epochs · lower MAE is better · ± is sample SD across seeds